

§2.4

P1. Solution:

$$y' = -y, y(0) = 2; y(x) = 2e^{-x} \quad (y(\frac{1}{2}) \approx 1.2131)$$

i) $h = 0.25, x_0 = 0, y_0 = 2, f(x, y) = -y$

$$x_1 = x_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 + (-y_0) \cdot h$$

$$= 2 + (-2) \cdot 0.25 = 1.5$$

$$x_2 = x_1 + h = 0.25 + 0.25 = 0.5$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h$$

$$= 1.5 + (-1.5) \cdot 0.25 = 1.125$$

$$|y_2 - y(\frac{1}{2})| \approx 0.0881$$

ii) $h = 0.1$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 + (-y_0) \cdot h$$

$$= 2 + (-2) \cdot (0.1) = 1.9$$

$$x_2 = x_1 + h = 0.2$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h$$

$$= 1.9 + (-1.9) \cdot (0.1) = 1.62$$

$$x_3 = x_2 + h = 0.3$$

$$y_3 = y_2 + f(x_2, y_2) \cdot h = 1.4580$$

$$x_4 = x_3 + h = 0.4$$

$$y_4 = y_3 + f(x_3, y_3) \cdot h = 1.3122$$

$$x_5 = x_4 + h = 0.5$$

$$y_5 = y_4 + f(x_4, y_4) \cdot h \approx 1.1810$$

$$|y_5 - y(\frac{1}{2})| \approx 0.0$$

P4. Solution:

$$y' = x - y, y(0) = 1; y(x) = 2e^{-x} + x - 1$$

$$(y(\frac{1}{2}) \approx 0.7131)$$

$$x_0 = 0, y_0 = 1, f(x, y) = x - y$$

i) $h = 0.25$

$$x_1 = x_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 + (x_0 - y_0) \cdot h$$

$$= 1 + (0 - 1) \cdot (0.25) = 0.75$$

$$x_2 = x_1 + h = 0.5$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h$$

$$= 0.75 + (0.25 - 0.75) \cdot (0.25) = 0.625$$

$$|y_2 - y(\frac{1}{2})| \approx 0.0881$$

ii) $h = 0.1$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = 1 + (0 - 1) \cdot (0.1) = 0.9$$

$$x_2 = x_1 + h = 0.2$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h = 0.9 + (0.1 - 0.9) \cdot 0.1$$

$$= 0.82$$

$$x_3 = x_2 + h = 0.3$$

$$y_3 = y_2 + f(x_2, y_2) \cdot h = 0.758$$

$$x_4 = x_3 + h = 0.4$$

$$y_4 = y_3 + f(x_3, y_3) \cdot h = 0.7122$$

$$x_5 = x_4 + h = 0.5$$

$$y_5 = y_4 + f(x_4, y_4) \cdot h \approx 0.6810$$

$$|y_5 - y(\frac{1}{2})| \approx 0.0321$$

P6. Solution:

$$y' = -2xy, y(0) = 2; y(x) = 2e^{-x^2} \quad (y(\frac{1}{2}) \approx 1.5576)$$

$$x_0 = 0, y_0 = 2, f(x, y) = -2xy$$

i) $h = 0.25$

$$x_1 = x_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 - 2 \cdot x_0 \cdot y_0 \cdot h$$

$$= 2 - 2 \cdot 0 \cdot 2 \cdot (0.25) = 2$$

$$x_2 = x_1 + h = 0.5$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h = 2 - 2 \cdot (0.25) \cdot 2 \cdot (0.25)$$

$$= 1.75$$

$$|y_2 - y(\frac{1}{2})| \approx 0.1924$$

ii) $h = 0.1$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 - 2 \cdot x_0 \cdot y_0 \cdot h$$

$$= 2 - 2 \cdot 0 \cdot 2 \cdot 0.1 = 2$$

$$x_2 = x_1 + h = 0.2$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h = 2 - 2 \cdot (0.1) \cdot 2 \cdot (0.1) = 1.96$$

$$x_3 = x_2 + h = 0.3$$

$$y_3 = y_2 + f(x_2, y_2) \cdot h = 1.8816$$

$$x_4 = x_3 + h = 0.4$$

$$y_4 = y_3 + f(x_3, y_3) \cdot h \approx 1.7687$$

$$x_5 = x_4 + h = 0.5$$

$$y_5 = y_4 + f(x_4, y_4) \cdot h \approx 1.6272$$

$$|y_5 - y(\frac{1}{2})| \approx 0.0696$$

$$x_5 = x_4 + h = 0.5$$

$$y_5 = y_4 + f(x_4, y_4) \cdot h \approx 0.498$$

$$|y_5 - y(\frac{1}{2})| \approx 0.0143$$

P8. solution:

$$y' = e^{-y}, y(0) = 0; y(x) = \ln(x+1), (y(\frac{1}{2}) \approx 0.4055)$$

$$x_0 = 0, y_0 = 0, f(x, y) = e^{-y}$$

i) $h = 0.25$

$$x_1 = x_0 + h = 0 + 0.25 = 0.25$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 + e^{-y_0} \cdot h$$

$$= 0 + e^{-0} \cdot (0.25) = 0.25$$

$$x_2 = 0.5$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h \approx 0.4447$$

$$|y_2 - y(\frac{1}{2})| \approx 0.0392$$

ii) $h = 0.1$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

$$y_1 = y_0 + f(x_0, y_0) \cdot h = y_0 + e^{-y_0} \cdot h$$

$$= 0 + e^{-0} \cdot 0.1 = 0.1$$

$$x_2 = x_1 + h = 0.1 + 0.1 = 0.2$$

$$y_2 = y_1 + f(x_1, y_1) \cdot h$$

$$= 0.1 + e^{(-0.1)} \cdot (0.1) \approx 0.1905$$

$$x_3 = x_2 + h = 0.3$$

$$y_3 = y_2 + f(x_2, y_2) \cdot h \approx 0.2731$$

$$x_4 = x_3 + h = 0.4$$

$$y_4 = y_3 + f(x_3, y_3) \cdot h \approx 0.3492$$

P20. Solution:

$$W(f, g) = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} \pi & \frac{\cos^2 x + \sin^2 x}{0} \\ 0 & 0 \end{vmatrix} = 0$$

$$= 0$$

$\Rightarrow$ linearly dependent

P22. Solution:

$$f(x) = 1+x, \quad g(x) = 1+|x|$$

$$f'(x) = 1, \quad g'(x) = \begin{cases} 1 & x > 0 \\ -1 & x < 0 \end{cases} \text{ not differentiable at } x=0.$$

$$\text{clearly } 1+x \neq C(1+|x|) \quad \forall x \in \mathbb{R}$$

for any C

$$\text{and } C(1+x) \neq 1+|x| \quad \forall x \in \mathbb{R}$$

for any C .

$\Rightarrow$ linearly independent.

P25. Solution:

$$f(x) = e^x \sin x, \quad g(x) = e^x \cos x$$

$$f'(x) = e^x \sin x + e^x \cos x$$

$$g'(x) = e^x \cos x - e^x \sin x$$

$$W(f, g) = \begin{vmatrix} e^x \sin x & e^x \cos x \\ e^x (\sin x + \cos x) & e^x (\cos x - \sin x) \end{vmatrix}$$

$$= e^{2x} (\sin x \cos x - \sin^2 x)$$

$$- e^{2x} (\sin x \cos x + \cos^2 x)$$

$$= -e^{2x} \neq 0$$

$\Rightarrow$ linearly independent.

P32. Solution:

$$\text{Characteristic equation: } r^2 - 3r + 2 = 0$$

$$\Rightarrow (r-2)(r-1) = 0$$

$$r_1 = 2, \quad r_2 = 1$$

$$\text{Thus } y_1(x) = e^{2x}, \quad y_2 = e^x$$

$$\Rightarrow y(x) = C_1 e^{2x} + C_2 e^x$$

P38. Solution:

$$\text{Characteristic equation: } 4r^2 + 8r + 3 = 0$$

$$\Rightarrow (2r+1)(2r+3) = 0$$

$$\Rightarrow r_1 = -\frac{1}{2}, \quad r_2 = -\frac{3}{2}$$

$$\text{Thus, } y_1 = e^{-\frac{1}{2}x}, \quad y_2 = e^{-\frac{3}{2}x}$$

$$\Rightarrow y(x) = C_1 e^{-\frac{1}{2}x} + C_2 e^{-\frac{3}{2}x}$$

P39. Solution:

$$\text{Characteristic equation: } 4r^2 + 4r + 1 = 0$$

$$\Rightarrow (2r+1)^2 = 0 \Rightarrow r_1 = r_2 = -\frac{1}{2}$$

$$\text{Thus, } y_1(x) = e^{-\frac{1}{2}x}, \quad y_2(x) = x e^{-\frac{1}{2}x}$$

$$\Rightarrow y(x) = C_1 e^{-\frac{1}{2}x} + C_2 x e^{-\frac{1}{2}x}$$

P40. Solution:

$$\text{Characteristic equation: } 9r^2 - 12r + 4 = 0$$

$$\Rightarrow (3r-2)^2 = 0 \Rightarrow r = \frac{2}{3}$$

$$\text{Thus, } y_1(x) = e^{\frac{2}{3}x}, \quad y_2(x) = x e^{\frac{2}{3}x}$$

$$\Rightarrow y(x) = C_1 e^{\frac{2}{3}x} + C_2 x e^{\frac{2}{3}x}$$

Ex 3.1.

P1. Solution:

$$y_1 = e^x, y_1' = e^x, y_1'' = e^x$$

$$\Rightarrow y_1'' - y_1 = 0$$

$$y_2 = e^{-x}, y_2' = -e^{-x}, y_2'' = e^{-x}$$

$$\Rightarrow y_2'' - y_2 = 0$$

$$y(x) = C_1 e^x + C_2 e^{-x} \Rightarrow$$

$$y'(x) = C_1 e^x - C_2 e^{-x} \Rightarrow$$

$$y(0) = C_1 + C_2 = 0 \Rightarrow \begin{cases} C_1 = 5/2 \\ C_2 = -5/2 \end{cases}$$

$$y'(0) = C_1 - C_2 = 5$$

$$\text{Thus, } y(x) = \frac{5}{2} e^x - \frac{5}{2} e^{-x}$$

P4. Solution:

$$y_1 = \cos 5x, y_1' = -5 \sin 5x, y_1'' = -25 \cos 5x$$

$$y_2 = \sin 5x, y_2' = 5 \cos 5x, y_2'' = -25 \sin 5x$$

$$\Rightarrow y_1'' + 25 y_1 = 0, y_2'' + 25 y_2 = 0$$

$$y(x) = C_1 \cos 5x + C_2 \sin 5x \Rightarrow$$

$$y'(x) = -5 C_1 \sin 5x + 5 C_2 \cos 5x$$

$$y(0) = C_1 = 10 \Rightarrow C_1 = 10$$

$$y'(0) = 5 C_2 = -10 \Rightarrow C_2 = -2$$

$$\text{Thus, } y(x) = 10 \cos 5x - 2 \sin 5x$$

P6. Solution:

$$y_1 = e^{2x}, y_1' = 2e^{2x}, y_1'' = 4e^{2x}$$

$$\Rightarrow y_1'' + y_1' - 6y_1 = 0$$

$$y_2 = e^{-3x}, y_2' = -3e^{-3x}, y_2'' = 9e^{-3x}$$

$$\Rightarrow y_2'' + y_2' - 6y_2 = 0$$

$$y(x) = C_1 e^{2x} + C_2 e^{-3x} \Rightarrow$$

$$y'(x) = 2C_1 e^{2x} - 3C_2 e^{-3x} \Rightarrow$$

$$\begin{cases} y(0) = C_1 + C_2 = 7 \\ y'(0) = 2C_1 - 3C_2 = -1 \end{cases} \Rightarrow \begin{cases} C_1 = 4 \\ C_2 = 3 \end{cases}$$

$$\text{Thus, } y(x) = 4e^{2x} + 3e^{-3x}$$

P9. Solution:

$$y_1 = e^{-x}, y_1' = -e^{-x}, y_1'' = e^{-x}$$

$$\Rightarrow y_1'' + 2y_1' + y_1 = 0$$

$$y_2 = xe^{-x}, y_2' = -xe^{-x} + e^{-x}, y_2'' = xe^{-x} - e^{-x} - e^{-x} = xe^{-x} - 2e^{-x}$$

$$\Rightarrow y_2'' + 2y_2' + y_2 = xe^{-x} - 2e^{-x} + 2(-xe^{-x} + e^{-x}) + xe^{-x} = 0$$

$$y(x) = C_1 e^{-x} + C_2 x e^{-x}$$

$$y'(x) = -C_1 e^{-x} + C_2 (e^{-x} - x e^{-x})$$

$$\Rightarrow y(0) = C_1 = 2 \Rightarrow \begin{cases} C_1 = 2 \\ C_2 = 1 \end{cases}$$

$$y'(0) = -C_1 + C_2 = -1$$

$$\text{Thus, } y(x) = 2e^{-x} + x e^{-x}$$

P12. Solution:

$$y_1 = e^{-3x} \cos 2x, y_1' = -3e^{-3x} \cos 2x - 2e^{-3x} \sin 2x$$

$$y_1'' = 9e^{-3x} \cos 2x + 6e^{-3x} \sin 2x + 6e^{-3x} \sin 2x - 4e^{-3x} \cos 2x$$

$$y_2 = e^{-3x} \sin 2x, y_2' = -3e^{-3x} \sin 2x + 2e^{-3x} \cos 2x$$

$$y_2'' = 9e^{-3x} \sin 2x - 6e^{-3x} \cos 2x - 6e^{-3x} \cos 2x - 4e^{-3x} \sin 2x$$

$$= -12e^{-3x} \cos 2x + 5e^{-3x} \sin 2x$$

$$\Rightarrow y_1'' + 6y_1' + 13y_1 = 5e^{-3x} \cos 2x + 12e^{-3x} \sin 2x + 6(-3e^{-3x} \cos 2x - 2e^{-3x} \sin 2x) + 13e^{-3x} \cos 2x$$

$$= 0$$

$$y_2'' + 6y_2' + 13y_2 = -12e^{-3x} \cos 2x + 5e^{-3x} \sin 2x + 6(-3e^{-3x} \sin 2x + 2e^{-3x} \cos 2x) + 13e^{-3x} \sin 2x$$

$$= 0$$

$$y(x) = C_1 e^{-3x} \cos 2x + C_2 e^{-3x} \sin 2x \quad \text{Thus } y(x) = 2e^{-3x} \cos 2x + 3e^{-3x} \sin 2x$$

$$y'(x) = C_1 (-3e^{-3x} \cos 2x - 2e^{-3x} \sin 2x) + C_2 (-3e^{-3x} \sin 2x + 2e^{-3x} \cos 2x)$$

$$\Rightarrow y(0) = C_1 = 2 \Rightarrow \begin{cases} C_1 = 2 \\ C_2 = 3 \end{cases}$$

$$\Rightarrow y'(0) = C_1 (-3) + C_2 (2) = 0 \Rightarrow C_2 = 3$$